

Para la función exponencial encontrar:

$$\sigma_x^2 = \int_0^{\infty} (x - \bar{x})^2 f_x(x) dx$$

$$E[(x - \mu)^2]$$

$$\sigma_x^2 = \int_0^{\infty} (x - \frac{1}{\lambda})^2 \lambda e^{-\lambda x} dx$$

$$\text{Vease que } \sigma_x^2 = E[x^2] - (E[x])^2$$

$$\Rightarrow E[x]^2 = \frac{1}{\lambda^2}$$

$$\Rightarrow E[x^2] = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx$$

Considerando que:

$$\int u dv = u \cdot v - \int v du$$

$$\Rightarrow u = (x - \frac{1}{\lambda})^2 \Rightarrow du = 2x dx$$

$$dv = \lambda e^{-\lambda x} \Rightarrow v = -e^{-\lambda x}$$

$$\int_0^{\infty} x^2 \lambda e^{-\lambda x} dx = \left[-x^2 e^{-\lambda x} \right]_0^{\infty} - \int_0^{\infty} -2x e^{-\lambda x} dx$$

$$= +2 \int_0^{\infty} x e^{-\lambda x} dx = \frac{2}{\lambda} \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

$$\therefore E[x^2] = \frac{2}{\lambda} \left(\frac{1}{\lambda} \right) = \frac{2}{\lambda^2}$$

$$\therefore \sigma_x^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

$$\therefore \sigma_x = \frac{1}{\lambda}$$

$$\leftarrow E[x] = \frac{1}{\lambda}$$

